

Clock Rank, Cycle Length, and Cloud Degree

An Exact Integer Translation and a Seven-Level Calibration Audit

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Abstract

The finite-cycle route introduces three integer scales: the RH-82 reset clock rank r_σ , the RH-15 effective cloud degree N_σ , and the cycle length $L_\sigma = N_\sigma + 1$. We derive their exact bookkeeping relation. Let

$$h_\sigma = \frac{\log(1/\sigma)}{2\log\lambda}, \quad r_\sigma = \lceil h_\sigma \rceil + 2, \quad d_\sigma = h_\sigma - N_\sigma.$$

Then

$$r_\sigma - L_\sigma = \lceil d_\sigma \rceil + 1.$$

Thus every bounded cloud-degree defect immediately becomes a bounded finite rank-to-cycle correction.

Joining the seven RH-15 cloud rows to the half-log clock gives $1.4456 \leq d_\sigma \leq 2.3376$ and therefore

$$r_\sigma - L_\sigma \in \{3, 4\}.$$

Four rows have gap three and three rows gap four. At the only scale shared with the certified RH-151 reset atlas, $\sigma = 0.01$, the actual reset rank is seven, the cloud degree is three, the cycle length is four, and the gap is three, exactly as the translation predicts.

The six smaller-noise ranks are formal RH-82 clock values, not observed RH-151 reset packets. One physical overlap cannot select a unique correction $L = r - 3$ versus $L = r - 4$, and seven cloud fits cannot prove an asymptotic degree law. The result supplies a precise calibration corridor, not a physical cycle theorem. A valid next step must derive the cycle from time-domain or transfer data and then test which integer branch survives.

1 Three clocks that must not be identified by notation

The prime-dynamics program now contains three different finite dimensions. First, RH-82 proposes the half-log source packet clock [5]. Second, RH-15 selects an effective outer cloud degree from the noisy transfer spectrum [1]. Third, RH-176–177 realize a degree- N geometric factor by a reduced cycle of length $L = N + 1$ [3, 4].

These dimensions are related but not equal by definition:

r_σ source-memory packet rank;

N_σ fitted degree per geometric cloud copy;

L_σ full cycle length, including the removed stationary mode.

Any physical bridge must explain their offsets rather than suppress them.

2 Half-log clock and exact translation

Fix the deterministic expansion parameter

$$\lambda = 1.6785735104283224$$

and define

$$h_\sigma = \frac{\log(1/\sigma)}{2 \log \lambda}. \quad (1)$$

The RH-82 clock rank with offset two is

$$r_\sigma = \lceil h_\sigma \rceil + 2. \quad (2)$$

For an observed cloud degree N_σ , define

$$L_\sigma = N_\sigma + 1, \quad d_\sigma = h_\sigma - N_\sigma. \quad (3)$$

Theorem 2.1 (Clock-cycle integer translation). *For every real $h > 0$ and integer $N \geq 0$, if $r = \lceil h \rceil + 2$, $L = N + 1$, and $d = h - N$, then*

$$r - L = \lceil d \rceil + 1. \quad (4)$$

More generally, a rank offset $a \geq 0$ gives $r - L = \lceil d \rceil + a - 1$.

Proof. Since N is an integer, $\lceil h \rceil - N = \lceil h - N \rceil = \lceil d \rceil$. Therefore

$$r - L = (\lceil h \rceil + 2) - (N + 1) = \lceil d \rceil + 1.$$

The offset- a formula is identical. □

Corollary 2.2 (Defect corridor). *If $1 < d \leq 2$, then $r - L = 3$. If $2 < d \leq 3$, then $r - L = 4$. Hence a cloud degree staying between one and three units below the half-log horizon yields a cycle length three or four units below the offset-two packet rank.*

The result is integer arithmetic, not a dynamical theorem. Its value is that every empirical degree discrepancy can now be translated without ambiguity.

3 Seven-level joined audit

The RH-15 cloud table contains seven noise scales. For each row we recompute h_σ from (1), read N_σ , set $L = N + 1$, and evaluate the formal clock rank (2).

σ	h_σ	N	L	formal r	$d = h - N$	$r - L$
10^{-2}	4.4456	3	4	7	1.4456	3
$4 \cdot 10^{-3}$	5.3302	3	4	8	2.3302	4
$2 \cdot 10^{-3}$	5.9993	4	5	8	1.9993	3
10^{-3}	6.6684	5	6	9	1.6684	3
$5 \cdot 10^{-4}$	7.3376	5	6	10	2.3376	4
$2 \cdot 10^{-4}$	8.2221	6	7	11	2.2221	4
10^{-4}	8.8912	7	8	11	1.8912	3

The archived column called “degree defect from half horizon” agrees with $d = h - N$ in every row. The exact translation identity has zero failures. The gap sequence is

$$3, 4, 3, 3, 4, 4, 3.$$

This bounded offset is more informative than a raw correlation: it states the precise finite integer ambiguity that a physical construction must resolve.

4 The sole physical reset overlap

The RH-151 reset atlas covers $\sigma = 0.16, 0.08, 0.04, 0.02, 0.01$ [2], while RH-15 covers 0.01 and smaller noise. Their intersection is the single scale $\sigma = 0.01$. At that scale:

$$h = 4.4456227, \quad r_{\text{reset}} = 7, \quad N_{\text{cloud}} = 3, \quad L_{\text{cycle}} = 4.$$

Thus

$$r_{\text{reset}} - L_{\text{cycle}} = 3,$$

and the actual reset rank agrees with the formal RH-82 clock.

This one overlap is a consistency check, not a fitted law. In particular, it cannot distinguish between a rule that usually uses gap three and one that switches between gaps three and four according to a secondary phase or parity condition.

5 Rounding sensitivity of the ceiling clock

The integer translation is exact once h_σ is fixed, but the map $h \mapsto \lceil h \rceil$ is discontinuous at integers. Define the ceiling margin

$$m_\sigma = \text{dist}(h_\sigma, \mathbb{Z}).$$

If an uncertainty interval for h_σ has radius smaller than m_σ , the formal rank is stable. If it crosses an integer, the clock rank and rank-cycle gap can change by one.

Among the seven rows, $\sigma = 0.002$ is exceptionally close to a boundary:

$$h_{0.002} = 5.9993011961, \quad m_{0.002} = 6.9880 \times 10^{-4}.$$

A perturbation of the half-clock larger than this can move the ceiling from six to seven and change the formal packet rank from eight to nine. The next smallest margin in the table is about 0.1088 at $\sigma = 10^{-4}$; the other rows are much less sensitive.

Proposition 5.1 (Interval-stable translation). *Let $h \in [h_-, h_+]$ and fix an integer degree N . If $\lceil h_- \rceil = \lceil h_+ \rceil$, then the clock rank and rank-cycle gap are constant throughout the interval. Otherwise the possible gaps are the consecutive integers*

$$\lceil h_- \rceil - N + 1, \dots, \lceil h_+ \rceil - N + 1.$$

Proof. The cycle length $N + 1$ is fixed, and the only varying integer is $\lceil h \rceil$. The ceiling takes every consecutive integer between its endpoint values. $\square$

For a validated all-level clock, uncertainty in the expansion parameter λ must therefore be propagated through h_σ before rounding. The current audit uses the fixed archived value exactly as specified and does not attach an interval to it.

6 What could generate the offset?

The integer difference has several possible structural sources:

1. the reset packet includes the explicit rank offset two used to protect weak modes, while the reduced cycle removes one stationary mode;
2. the cloud degree is selected by a radial/phase threshold and can lag the half-log horizon by one additional mode;
3. left and right packet channels may share or split stationary and parity directions differently from the two cloud copies;
4. finite-resolution eigensolvers may undercount a mode near the separating radius.

The arithmetic theorem does not choose among these mechanisms. A physical cycle construction must expose which dimensions are stationary, protected, or genuinely resonant.

7 Two calibration branches

The observed corridor suggests two finite hypotheses:

$$\mathbf{C}_3 : L_\sigma = r_\sigma - 3, \tag{5}$$

$$\mathbf{C}_4 : L_\sigma = r_\sigma - 4. \tag{6}$$

Neither is asserted globally. The current seven rows select $\mathbf{C}_3$ four times and $\mathbf{C}_4$ three times. A third possibility is a switching law determined by the fractional part of h_σ or by an independently measured boundary phase.

8 Total mode count and two falsifiable branches

The doubled reduced cycle has total nonstationary rank

$$m_\sigma = 2N_\sigma = 2(L_\sigma - 1). \tag{7}$$

Combining this with a fixed rank-cycle gap $g = r - L$ gives an exact conversion from reset rank to cloud count.

Proposition 8.1 (Mode counts under the two calibration branches). *If $L = r - 3$, then*

$$m = 2r - 8. \quad (8)$$

If $L = r - 4$, then

$$m = 2r - 10. \quad (9)$$

At the physical overlap $r = 7$, the observed cloud count six selects the first formula and rejects the second at that scale.

Proof. Substitute $N = L - 1 = r - g - 1$ into $m = 2N$. For $g = 3$ and $g = 4$ this gives (8) and (9). When $r = 7$, the values are six and four, respectively; RH-15 records six cloud eigenvalues. $\square$

This is stronger than comparing only L and r : it predicts the full doubled cloud multiplicity. At a future scale where both a certified reset rank and a validated Riesz cloud count are available, the two branches can be tested without fitting phases or radii. If neither count matches, the simple fixed offset model is false even if the half-log and cloud degree remain correlated.

The current seven formal rows give predicted total counts

$$6, 6, 8, 10, 10, 12, 14,$$

exactly the archived values $2N$. This agreement is tautological on the cloud side because $m = 2N$ defines the model count; only the single reset overlap tests a cross-family prediction.

9 Falsification criteria for the calibration corridor

The finite corridor should be abandoned or revised if any of the following occurs:

1. a validated cloud degree has $h - N \notin (1, 3]$ persistently, producing a gap outside $\{3, 4\}$;
2. two or more scales with certified reset packets disagree with the formal clock rank after interval propagation;
3. the physical Riesz cloud has a mode count different from $2N$ because one copy is absent, extra stationary modes survive, or multiplicities split;
4. a time-domain cyclic construction selects a length incompatible with both $r - 3$ and $r - 4$.

These are clean negative outcomes. The purpose of the integer dictionary is to make the cycle hypothesis testable before a large determinant calculation, not to protect it from contradictory data.

Proposition 9.1 (What would make calibration rigorous). *Suppose an all-level cloud theorem proves integers N_σ satisfying*

$$a < h_\sigma - N_\sigma \leq b$$

for fixed real a, b . Then the rank-cycle gap lies in the finite set

$$\{[a] + 1, [a] + 2, \dots, [b] + 1\},$$

with endpoint adjustments when a is an integer. If the defect eventually lies in one unit interval $(k - 1, k]$, then the gap is eventually the single integer $k + 1$.

Proof. Apply Theorem 2.1 and monotonicity of the ceiling function. $\square$

Thus the needed analytic target is not exact equality of two ranks. It is an eventual unit-width bound on the cloud-degree defect.

10 Audit boundary

The joined script reads the archived RH-15 CSV and RH-151 JSON, recomputes the RH-82 half-log clock, and checks the integer identity. It has seven rows, one actual reset overlap, zero translation failures, and zero reset-clock mismatches on that overlap.

The six smaller-noise values $r = 8, 8, 9, 10, 11, 11$ are predictions of the clock formula only. No reset packets were computed or certified at those six scales in this paper. Likewise, the RH-15 values $N = 3, 4, 5, 6, 7$ are numerical cloud selections, not an all-level theorem.

11 Route consequence

The finite-cycle branch now has an exact integer dictionary and a narrow empirical corridor:

$$\text{cloud degree } N \leftrightarrow \text{cycle length } L = N + 1, \quad r - L \in \{3, 4\} \text{ on seven rows.}$$

The next step cannot be more bookkeeping. It must place the cycle operator inside, or close to, a physical transfer-space block and use explicit root spacing to construct Riesz shells. RH-180 gives that conditional theorem.

No unique cycle calibration, asymptotic degree law, physical Riesz cloud, Gate-A determinant, Hilbert–Polya operator, zeta identity, or Riemann Hypothesis is proved.

References

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